

Methods

0.1 MRI acquisition and preprocessing

MR images were acquired on a 3T whole-body MRI system (Magnetom Prisma, Siemens Medical Systems, Germany) using an 80 mT/m gradient coil and a 64-channel radio-frequency receive head coil at the Wolfson Brain Imaging Centre, Cambridge.

3D MP-RAGE images (Magnetisation-Prepared RAPid Gradient Echo) were acquired with isotropic voxel size = 1.0 mm^3 , TR = 2000 ms, TE = 1.95 ms, TI = 880 ms, flip angle = 8° , FOV = $256 \times 256 \times 208$ mm, 208 sagittal slices, GRAPPA reconstruction and acceleration factor = 2).

FLAIR images (FLuid-Attenuated Inversion Recovery) were acquired with voxel size $1.0 \times 1.0 \times 1.05$ mm, TR = 5000 ms, TE = 386 ms, TI = 1800 ms, FOV = $256 \times 256 \times 202$ mm, 192 sagittal slices, GRAPPA reconstruction and acceleration factor = 3, partial Fourier factor = $7/8$.

Multi-shell diffusion MR images were acquired with a single-shot SE-EPI sequence, TR = 2433 ms, TE = 75.60 ms, in-plane spatial resolution of $1.7 \times 1.7 \text{ mm}^2$, 76 slices with thickness of 1.8 mm, multi-band factor 4, no Partial Fourier encoding or GRAPPA acceleration; diffusion-weighted images were acquired with b-values of 300, 1000 and 2000 s/mm^2 in 8, 30 and 60 gradient directions uniformly distributed on a sphere, respectively; all diffusion-weighted images were acquired with opposing phase-encoding in order to perform correction for susceptibility-induced geometrical distortion; additionally, 6 non-diffusion-weighted images were acquired. The total acquisition time was 18:22 minutes.

0.1.1 Structural MRI and WMH segmentation

For WMH segmentation, FSL's TrUE-Net toolbox was used which is a deep-learning tool that utilises the spatial distribution of WMH in the loss functions during optimisation (Triplanar U-Net ensemble network model; Sundaresan, Zamboni, Rothwell, et al., 2021; Sundaresan, Zamboni, Dinsdale, et al., 2021; Sundaresan et al., 2022). T1-weighted and FLAIR images were preprocessed with TrUE-Net's built-in function `prepare_truenet_data` which performs skull-stripping, bias field correction and a linear rigid-body registration of the T1-weighted image to the FLAIR image. WMH were then segmented using TrUE-Net's model that was pretrained on MICCAI WMH Segmentation Challenge Data. Resulting maps were thresholded at 0.05 and masked with subject-specific WM masks generated by `fsl_nat` to remove false positives outside of the white matter. Maps then underwent visual inspection and comparison with overlaid FLAIR images. Some previous studies have excluded WMH voxels in caps around the frontal

horns of ventricles and ventricular linings (e.g. Enzinger et al., 2006) because it has been suggested that these could arise due to altered periventricular fluid dynamics as opposed to the ischemic tissue damage causing irregular periventricular WMH and WMH in the deep white matter (Fazekas et al., 1993). Here we did not exclude any WMH voxels in order to not risk arbitrary cut-offs, especially given that periventricular WMH which develop irregular extensions into the deep white matter do represent pathological entities ("Caps and a smooth halo of PVH should remain to be analyzed separately, while irregular PVH and confluent DWMH should be treated together", Fazekas et al., 1993).

0.1.2 dMRI

Diffusion MRI data was preprocessed with Micapipe version 0.2.3 (Cruces et al., 2022) and default settings, including the steps denoising to remove thermal noise, Gibbs ring correction, distortion and motion correction, and B1 bias field correction. Additional preprocessing steps whose benefits have previously been reported but which are not readily available within toolboxes were not undertaken in order to keep the pipeline as reproducible as possible. This concerns in particular Rician/noncentral-Chi bias correction, gradient nonlinearity correction, signal drift correction and signal outlier detection/correction (ref).

0.1.3 NODDI

The NODDI model (Neurite Orientation Dispersion and Density Imaging; Zhang et al., 2012) was fit to preprocessed dMRI images with the AMICO implementation (Accelerated Microstructure Imaging via Convex Optimization; Daducci et al., 2015), estimating the three parameters Neurite Density Index (NDI), Orientation Dispersion Index (ODI) and Free Water Fraction (FWF). The default parameters from Zhang et al., 2012 were used for parallel diffusivity $d = 1.7 \times 10^3 \text{mm}^2 \text{s}^{-1}$ and isotropic diffusivity $d_{iso} = 3.0 \times 10^3 \text{mm}^2 \text{s}^{-1}$.

0.1.4 fwDTI

The fwDTI model was fit to preprocessed dMRI data with DIPY (Garyfallidis et al., 2014; implementation described in Henriques et al., 2017, using a version of the free water elimination suggested in Hoy et al., 2014). Images collected at the highest b-value of 2000 s/mm^2 were excluded from the fwDTI analysis as the assumption of Gaussian diffusion underlying the tensor model does not hold at higher b-values. Maps of extracellular free water (FW) as well as FW-corrected maps of fractional anisotropy (FA), mean

diffusivity (MD), axial diffusivity (AD) and radial diffusivity (RD) were extracted. MD, AD and RD maps were thresholded so that values above 0.002 were removed in order to remove artificially high values. Additionally, voxels with high water volume fraction above the threshold of 0.7 (i.e. voxels containing mostly CSF) were excluded on FA maps in order to mitigate the known artefact of unexpectedly high FA values in these regions.

0.2 MRI analysis

Figure 1 gives an overview of subsequent processing and analysis steps described in the following.

0.2.1 Registration of WMH masks to dMRI images and extraction of dMRI metrics within WMH and NAWM

Each participant's FLAIR image was registered to their non-diffusion-weighted image with FSL FLIRT (Jenkinson et al., 2002; Jenkinson and Smith, 2001; Greve and Fischl, 2009), the transformation was applied to each participant's WMH mask, thus registering it to the dMRI space. WMH masks were then applied to all NODDI and fwDTI maps in order to sample and average the NDI, ODI, FWF, FA, MD, AD, RD and FW values across WMH voxels. Similarly, values of each dMRI metric were sampled and averaged from NAWM, i.e. all voxels within a participant's WM mask that were not part of WMH.

0.2.2 Construction of healthy WM reference maps

In order to compare WMH-averaged microstructure metrics with metrics in healthy white matter, we employed two strategies. Firstly, we used metrics averaged across NAWM voxels. However, this approach bears a risk of systematic bias since WMH predominantly occur in strategic regions of the brain instead of being uniformly distributed (Enzinger et al., 2006). Differences in microstructural metrics between WMH and NAWM could thus reflect an effect of location and not conclusively an effect of tissue damage. Therefore, ideally WMH need to be compared to healthy white matter in the same location, which we implemented with our second strategy, namely by creating reference dMRI maps of healthy white matter.¹ We identified reference participants with no visible white matter changes on FLAIR MRI and stratified them by sex and age such that each group had a mean age close to target ages (40-85 years in 5-year increments). For each age- and sex-specific group, a group-average template was created using

¹Previous studies have compared WMH tissue properties to those of contralateral healthy WM. Given the often symmetrical distribution of WMH in our population which made it difficult to find healthy contralateral tissue, we opted for a different approach.

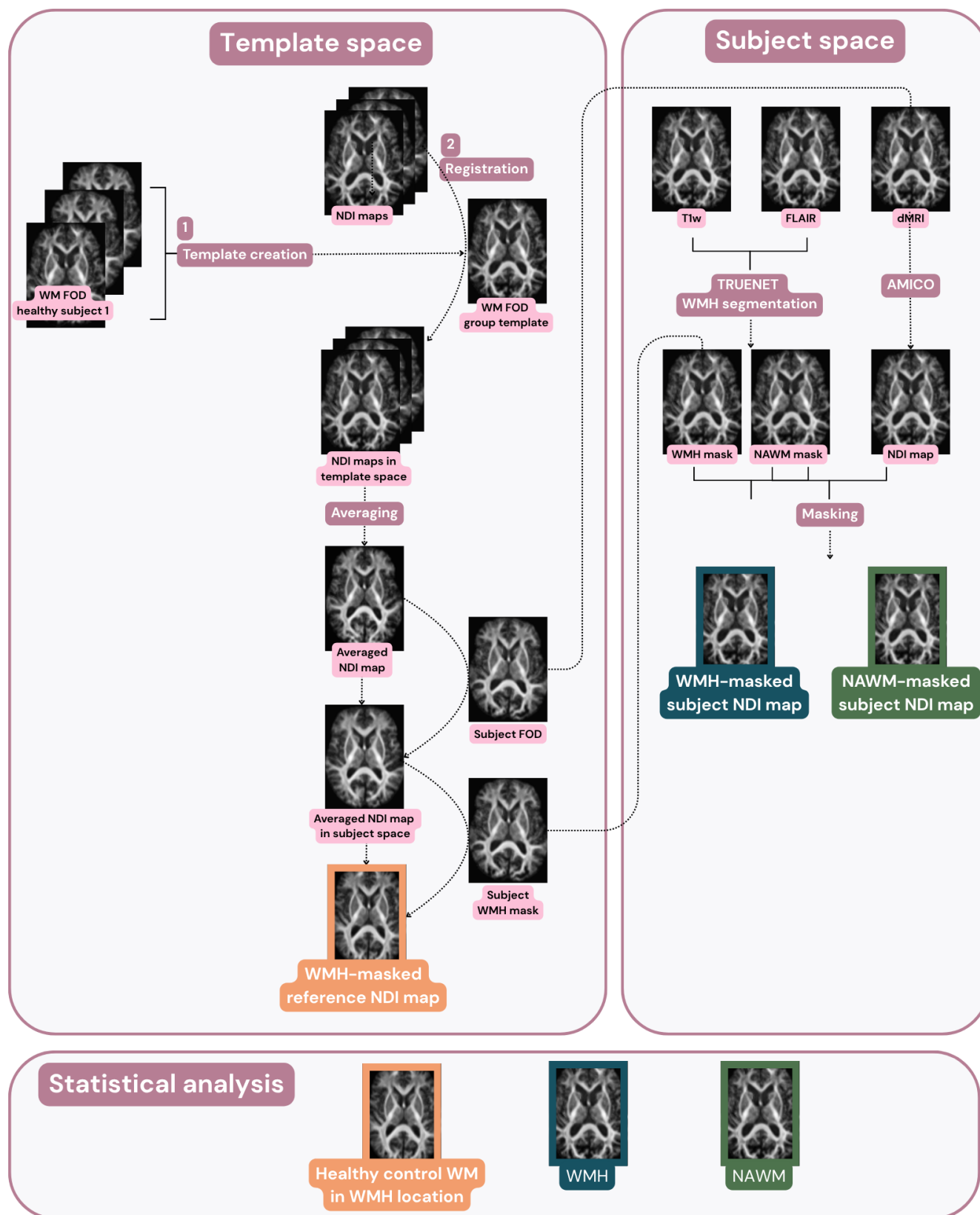


Figure 1: MRI analysis pipeline. 19 age- and sex-specific templates were created from the images of participants with no WMH. For each template, all dMRI maps were generated. These reference images served as normal controls to be compared to images with WMH.

WM FODs and an iterative registration and averaging approach (Raffelt et al., 2011, implemented in the MRtrix3 command *population_ttemplate*)². For each age- and sex-specific group, each reference participant's WM FOD image was then nonlinearly registered to the group template (Raffelt et al., 2011, implemented in the MRtrix3 command *mrregister*), and the resulting transformations were applied to the participant's dMRI maps (NDI, ODI, FWF, FA, MD, AD, RD, and FW) to warp them to template space. These dMRI maps in template space were averaged, resulting in age- and sex-specific reference maps of healthy white matter. Finally, each study participant was assigned one set of these reference maps so that the participant's age matched the reference age as closely as possible. Reference maps were registered to each study participant's native diffusion space and were masked with the template CSF mask. This was important as many WMH voxels are located at the ventricle border and as template registration is not perfect enough to avoid WMH voxels being registered to CSF voxels] The mean across all WMH voxels in reference maps was calculated and represented an estimate of healthy white matter at anatomical locations matching the locations of WMH.

0.3 Statistical analysis

All analyses were conducted in Python (version 3.14.1). False discovery rate (FDR; Benjamini and Hochberg, 1995) adjustment was applied where appropriate to correct resulting p -values for multiple comparisons.

0.3.1 Microstructural signature of WMH and NAWM in individuals with white matter pathology

In order to test whether tissue microstructure differed between WMH and normal WM in matched locations, Welch's t -test was performed for all NODDI and fwDTI metrics. This analysis was repeated for participants' normal-appearing white matter (NAWM) instead of WMH. The aim was to examine whether NAWM had also undergone microstructural changes despite its unassuming appearance on structural images, and to understand whether changes in WMH were specific to these lesioned areas or were developments present in the entire white matter.

²WM FODs were selected for template construction because they resulted in higher-quality templates than non-diffusion-weighted ($b = 0$) images.

0.3.2 WMH microstructure in the context of morphology

Visual inspection suggested a non-linear relationship between microstructure metrics and WMH volume/confluence, consistent with a log-linear association. Volume and confluence scores were therefore log-transformed before entering linear regressions. To investigate the relationship between WMH microstructure and WMH morphology (i.e., volume and confluence), separate multiple regressions of each WMH-averaged NODDI and fwDTI metric against WMH volume and confluence, age and sex were conducted. This analysis was also repeated for NAWM instead of WMH in order to understand whether the extent of invisible changes in NAWM depended on WMH burden.

0.3.3 WMH microstructure in the context of location

In order to study the association between WMH microstructure and the distance to the ventricles, voxelwise Pearson correlations of each NODDI and fwDTI metric and the voxel's distance to the ventricles were calculated.

0.3.4 Joint analysis of WMH morphology and location

In order to address the question whether high- and low-confluence lesions predominantly occur in distinct locations, lesion probability maps were created based on WMH maps for the top and bottom quartiles of confluence (i.e. the 25% of participants with the highest and lowest confluence scores respectively). Firstly, all WMH maps had to be registered to a common template space. For this, from participants with the lowest WMH burdens we selected 20 participants such that their mean age matched the mean age of the overall cohort (72 years). A group template was created from their WM FOD images using the procedure described above. FLAIR images from all remaining participants were registered to the group template in a two-step process via subject-space dMRI images (linear registration of FLAIR images to dMRI images and subsequent non-linear registration of dMRI images to group template); the transforms were then applied to the size-controlled WMH masks described above in order to transform them to group template space. WMH masks for the top and bottom quartiles were then summed and the resulting images thresholded at a value of 5, i.e. only voxels that belonged to WMH masks in 5 or more participants remained in the lesion probability map. Since high-confluence lesions occupy a larger area than low-confluence lesions, the difference in the number of voxels over which dMRI metrics are averaged could introduce a bias into the lesion probability maps. This caveat was already noted in Enzinger et al., 2006 and should therefore be mitigated. Therefore for this analysis in particular we did not use regular

WMH masks, but instead created size-controlled WMH masks. For this, we determined the centre of mass of each WMH cluster (in axial 2D slices) and generated circular WMH with a radius of 3 voxels around each centre of mass. This ensured that the location of WMH clusters remained the same while the size was controlled and would not introduce bias into the lesion probability maps. Resulting maps were visually inspected and the spatial distribution of lesions reported.

While the previous analysis addressed whether high- and low-confluence lesions predominantly occur in distinct *locations* in the brain, we next investigated whether these lesions preferentially arise within specific *microstructural environments* that differ already in healthy white matter [maybe better "arise in white matter regions with distinct baseline microstructural properties"?]. To this end, the linear regressions of NODDI and fwDTI metrics against confluence scores were repeated using healthy white matter sampled from WMH locations. Specifically, the size-controlled WMH maps described above were transformed into healthy participant space and used as masks to sample dMRI metrics from normal white matter at locations corresponding to WMH observed in the patient cohort. Thus, the predictor variable remained the WMH confluence score derived from participants with WMH, whereas the dependent variables were microstructural metrics measured in healthy tissue at matched anatomical locations. Size-controlled WMH maps were again used in order to mitigate potential bias arising from differences in WMH size.

References

- Benjamini, Y., & Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society. Series B (Methodological)*, 57(1), 289–300. Retrieved July 25, 2025, from <https://www.jstor.org/stable/2346101>
- Cruces, R. R., Royer, J., Herholz, P., Larivière, S., Vos de Wael, R., Paquola, C., Benkarim, O., Park, B.-Y., Degré-Pelletier, J., Nelson, M. C., DeKraker, J., Leppert, I. R., Tardif, C., Poline, J.-B., Concha, L., & Bernhardt, B. C. (2022). Micapipe: A pipeline for multimodal neuroimaging and connectome analysis. *NeuroImage*, 263, 119612. <https://doi.org/10.1016/j.neuroimage.2022.119612>
- Daducci, A., Canales-Rodríguez, E. J., Zhang, H., Dyrby, T. B., Alexander, D. C., & Thiran, J.-P. (2015). Accelerated Microstructure Imaging via Convex Optimization (AMICO) from diffusion MRI data. *NeuroImage*, 105, 32–44. <https://doi.org/10.1016/j.neuroimage.2014.10.026>
- Enzinger, C., Smith, S., Fazekas, F., Drevin, G., Ropele, S., Nichols, T., Behrens, T., Schmidt, R., & Matthews, P. M. (2006). Lesion probability maps of white matter hyperintensities in elderly individuals. *Journal of Neurology*, 253(8), 1064–1070. <https://doi.org/10.1007/s00415-006-0164-5>
- Fazekas, F., Kleinert, R., Offenbacher, H., Schmidt, R., Kleinert, G., Payer, F., Radner, H., & Lechner, H. (1993). Pathologic correlates of incidental MRI white matter signal hyperintensities. *Neurology*, 43(9), 1683–1683. <https://doi.org/10.1212/WNL.43.9.1683>
- Garyfallidis, E., Brett, M., Amirbekian, B., Rokem, A., van der Walt, S., Descoteaux, M., Nimmo-Smith, I., & Dipy Contributors. (2014). Dipy, a library for the analysis of diffusion MRI data. *Frontiers in Neuroinformatics*, 8, 8. <https://doi.org/10.3389/fninf.2014.00008>
- Greve, D. N., & Fischl, B. (2009). Accurate and robust brain image alignment using boundary-based registration. *NeuroImage*, 48(1), 63–72. <https://doi.org/10.1016/j.neuroimage.2009.06.060>
- Henriques, R. N., Rokem, A., Garyfallidis, E., St-Jean, S., Peterson, E. T., & Correia, M. M. (2017, February). [Re] Optimization of a free water elimination two-compartment model for diffusion tensor imaging [Pages: 108795 Section: Confirmatory Results]. <https://doi.org/10.1101/108795>
- Hoy, A. R., Koay, C. G., Kecskesti, S. R., & Alexander, A. L. (2014). Optimization of a free water elimination two-compartment model for diffusion tensor imaging. *NeuroImage*, 103, 323–333. <https://doi.org/10.1016/j.neuroimage.2014.09.053>
- Jenkinson, M., Bannister, P., Brady, M., & Smith, S. (2002). Improved Optimization for the Robust and Accurate Linear Registration and Motion Correction of Brain Images. *NeuroImage*, 17(2), 825–841. <https://doi.org/10.1006/nimg.2002.1132>

- Jenkinson, M., & Smith, S. (2001). A global optimisation method for robust affine registration of brain images. *Medical Image Analysis*, 5(2), 143–156. [https://doi.org/10.1016/S1361-8415\(01\)00036-6](https://doi.org/10.1016/S1361-8415(01)00036-6)
- Raffelt, D., Tournier, J.-D., Frupp, J., Crozier, S., Connelly, A., & Salvado, O. (2011). Symmetric diffeomorphic registration of fibre orientation distributions. *NeuroImage*, 56(3), 1171–1180. <https://doi.org/10.1016/j.neuroimage.2011.02.014>
- Sundaresan, V., Dinsdale, N. K., Jenkinson, M., & Griffanti, L. (2022). *Omni-supervised domain adversarial training for white matter hyperintensity segmentation in the UK Biobank*. IEEE. Retrieved September 25, 2025, from <https://ora.ox.ac.uk/objects/uuid:780c327d-9ce2-4007-9baa-7233e1d9cf54>
- Sundaresan, V., Zamboni, G., Dinsdale, N. K., Rothwell, P. M., Griffanti, L., & Jenkinson, M. (2021). Comparison of domain adaptation techniques for white matter hyperintensity segmentation in brain MR images. *Medical Image Analysis*, 74, 102215. <https://doi.org/10.1016/j.media.2021.102215>
- Sundaresan, V., Zamboni, G., Rothwell, P. M., Jenkinson, M., & Griffanti, L. (2021). Triplanar ensemble U-Net model for white matter hyperintensities segmentation on MR images. *Medical Image Analysis*, 73, 102184. <https://doi.org/10.1016/j.media.2021.102184>
- Zhang, H., Schneider, T., Wheeler-Kingshott, C. A., & Alexander, D. C. (2012). NODDI: Practical in vivo neurite orientation dispersion and density imaging of the human brain. *NeuroImage*, 61(4), 1000–1016. <https://doi.org/10.1016/j.neuroimage.2012.03.072>